Jos Buttler Runs Prediction

AI Course Project Proposal | FUUAST, Karachi

Submitted by: Asif Siddique

1. Selected Player

Jos Buttler (English cricketer, right-handed wicketkeeper-batter)

2. Objective

Predict how many runs Jos Buttler will score in his next international innings (ODI, T20I or Test).

3. Approach

Supervised learning (the output is a numerical value: runs scored). This is a regression problem.

4. Model

Decision Tree Regressor (main model), compared with Random Forest Regressor.

5. Reason

- Supervised learning: each previous international innings has a known runs-scored value, so the data is labelled.
- Decision Tree: can use factors such as opponent, venue, batting position and recent form; it can model non-linear patterns and is relatively easy to interpret.
- Random Forest: combines multiple decision trees, which can reduce overfitting and provide more stable predictions.

6. Dataset (Authentic Sources)

Main source (ICC): Jos Buttler player profile and rankings
https://www.icc-cricket.com/rankings/9782/jos-buttler

Verification source (ESPNcricinfo Statsguru): International cricket batting statistics and match records
https://stats.cricinfo.com/cricinfo/engine/stats/index.html

Additional source (ESPNcricinfo player profile): Player profile, role and career summary
https://www.espncricinfo.com/player/jos-buttler-308967

The dataset (Asif_Fg_Jos_Butler_.csv) contains 422 international innings from 25 September 2011 to 24 September 2026 (ODI: 174, T20I: 148, Test: 100). Columns include Match_Date, Opposition, Venue, Batting_Position, Runs, Balls_Faced, Fours, Sixes, Strike_Rate, Previous_Match_Runs, Last_3_Match_Avg, Last_5_Match_Avg and Format. The target column is Runs (runs scored in each innings). Records are checked against the ICC and ESPNcricinfo sources before model training.

RESULT SCREENSHOTS

Altair AI Studio (Trial 2026.1.1)

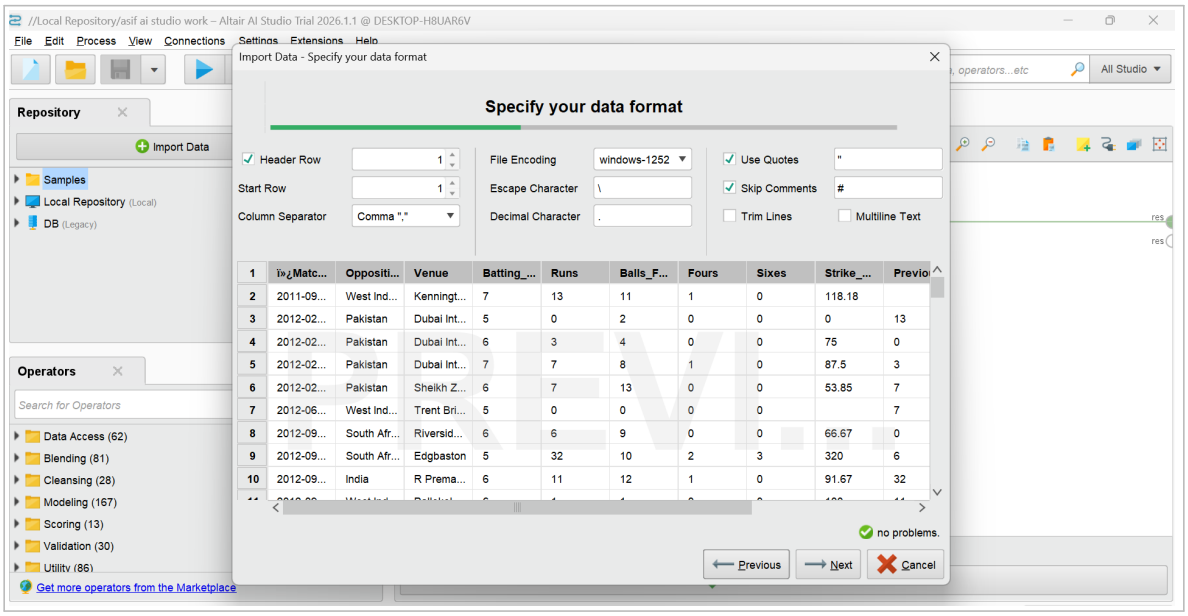


Figure 1: Importing the dataset

The Jos Buttler innings data file is loaded using the Import Data wizard (header row enabled, comma separator, windows-1252 encoding). The preview shows columns such as Match date, Opposition, Venue, Batting position, Runs, Balls Faced, Fours, Sixes and Strike Rate.

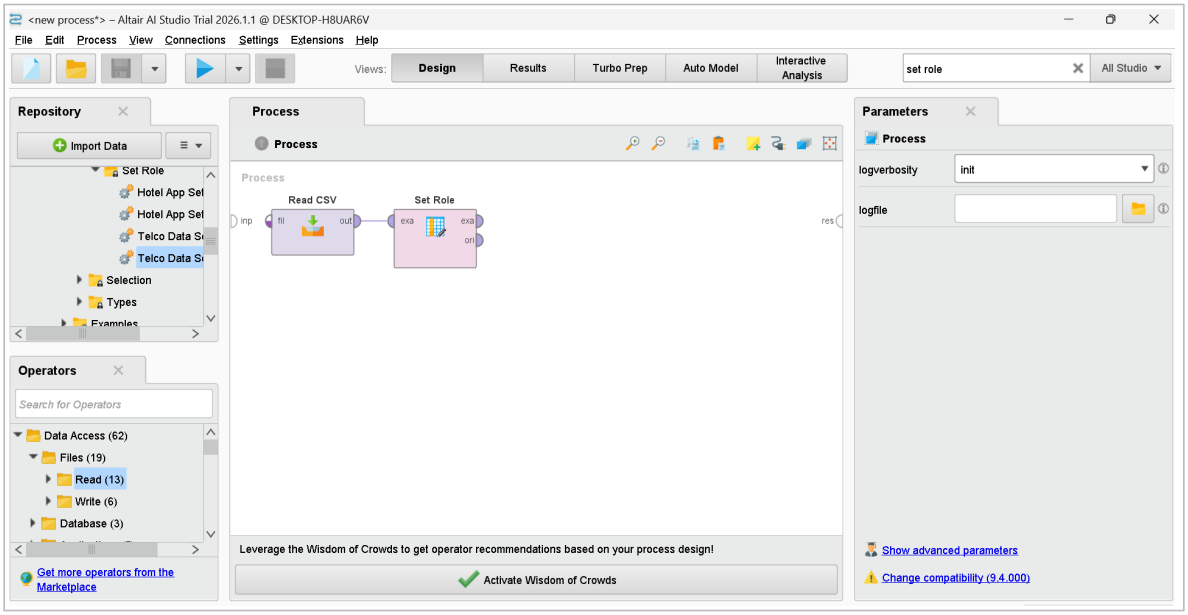


Figure 2: Read CSV and Set Role

The Read CSV operator loads the data, and the Set Role operator is connected to it so the target (label) column, runs scored, can be defined for supervised learning.

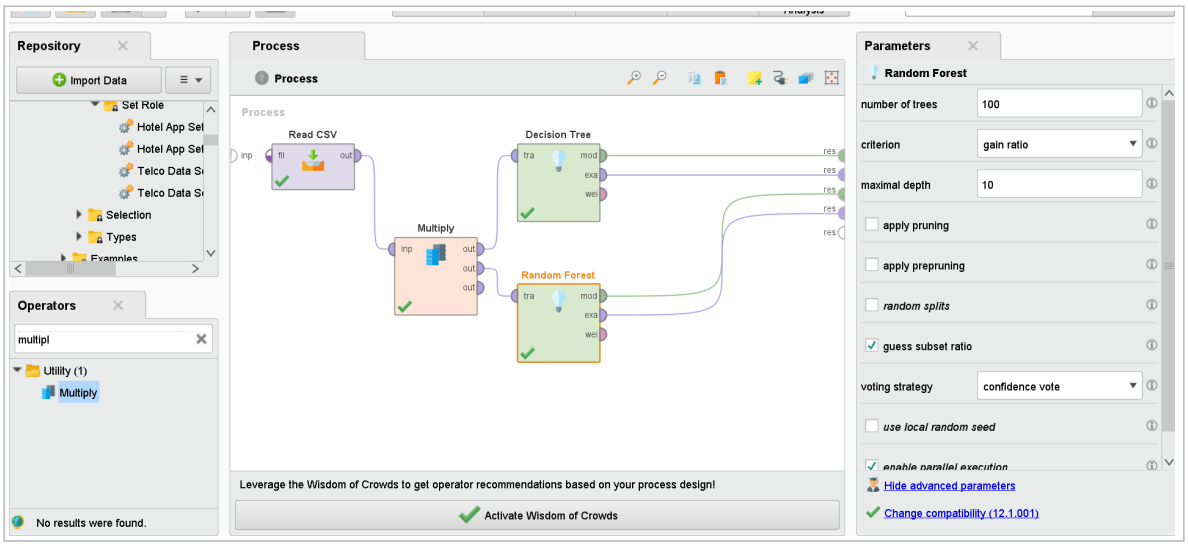


Figure 3: Process design with Multiply, Decision Tree and Random Forest

The loaded data passes through a Multiply operator so the same dataset feeds both the Decision Tree and the Random Forest (100 trees, criterion: gain ratio, maximal depth: 10), allowing the two models to be compared.

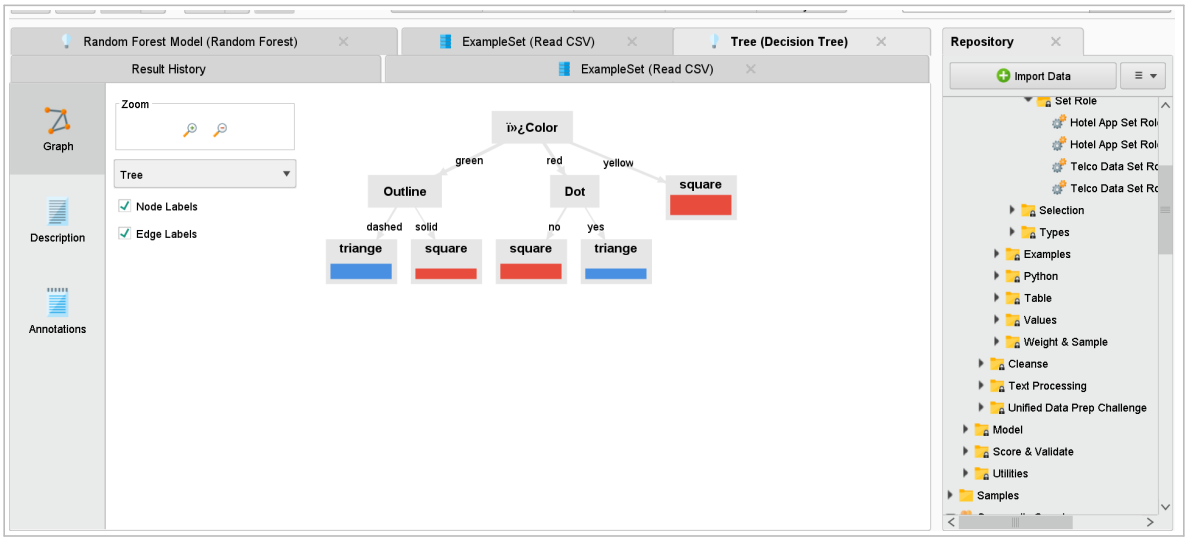


Figure 4: Decision Tree result

Graph view of the trained Decision Tree model, showing the splits (nodes), edge labels and the predictions at the leaves.

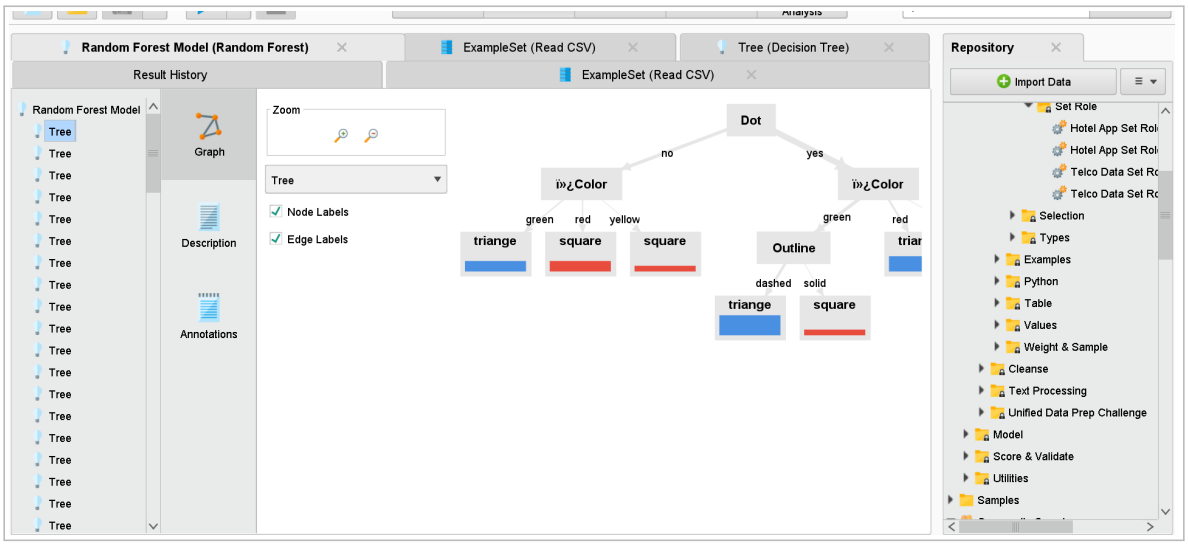


Figure 5: Random Forest model result

The Random Forest model is made up of many individual trees; the left panel lists each tree and the graph shows the selected tree's structure.